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At XYZ Inc. any employee receives a verbal warning upon accumulating at least 3 unexcused absences within any 365-day period and a written reprimand upon accumulating at least 4 such absences. For any single 8-hour workday, missing between 10 minutes and 2 hours of work counts as one-third of an absence, missing between 2 hours and 4 hours of work counts as half an absence, and missing more than 4 hours counts as a full absence. However, an employee may stay late to make up for up to 1 hour of an unexcused absence on the same day.

The following options are descriptions of the unexcused absences of 5 employees of XYZ Inc. Assume that in each case the employee had no other unexcused absences and made up no other time. Select a description of an employee who qualified for a verbal warning but not a written reprimand, and select a description of an employee who qualified for a written reprimand. Make only two selections, one in each column.

| **Verbal warning** | **Written reprimand** | |
|---|---|---|
| | | Absent all day on 5 April 2010, 8 June 2010, 17 April 2011, and 14 June 2011 |
| | | Absent 4.5 hours but stayed 1 hour late on 13 May 2010; absent all day on 2 June 2010, 1 May 2011, and 21 July 2011 |
| | | Absent 4.5 hours on 19 March 2010; stayed one hour late on 20 March 2010; absent all day on 8 February 2011 and 9 February 2011; arrived 40 minutes late on 17 April 2011 |
| | | Absent 3.5 hours on 13 September 2010; absent 1 hour on 15 September 2010; absent 6 hours on 16 September 2010; absent 2.5 hours on 18 September 2010; absent 1 hour on 19 September 2010 |
| | | Absent 3 hours on 7 July 2010; absent 2.5 hours on 13 September 2010; absent all day on 31 January 2011 and 4 July 2011; absent 5 hours on 12 March 2011 |

•

Marketing strategist: Agency A designed an advertising campaign that our company is about to test with a focus group. We are wondering whether a new ad campaign will increase our name recognition among consumers. As a contingency, we have decided that we might ask Agency B to design an alternate campaign. However, if we find that Agency A's campaign elicits positive responses from the focus group, we will not ask Agency B for a campaign.

A statement that must be true if the marketing strategist's statements are true: After the focus testing is complete, if the company ___1___, then it must also be the case that the company ___2___.

Select for 1 *if* and 2 *then* the two different options that create a statement that must be true if the marketing strategist's statements are true. Make only two selections, one in each column.

| **1 if** | **2 then** | |
|---|---|---|
| | | believes Agency A's campaign will increase the company's name recognition |
| | | believes Agency A's campaign will decrease the company's name recognition |
| | | asks Agency B for a campaign |
| | | asks Agency A for an alternate campaign |
| | | believes the focus group did not respond positively to Agency A's campaign |

•

Francois and Pierre each owe Claudine money. Today, Francois will make a payment equal to 50% of the amount he owes Claudine, and Pierre will make a payment equal to 10% of the amount he owes Claudine. Together, the two payments will be equal to 40% of the combined amount that Francois and Pierre owe Claudine.

Select for *Francois* and *Pierre* amounts that Francois and Pierre could owe Claudine that are jointly consistent with the given information. Make only two selections, one in each column.

| **Francois** | **Pierre** | |
|---|---|---|
| | | €50 |
| | | €250 |
| | | €750 |

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| | | |
|---|---|---|
| | | €750 |
| | | €3,750 |
| | | €6,750 |

•

Witness testimony: Around 8:00 on the morning of Tuesday, July 6, I saw a man exit from the back entrance of the building. I knew it was McGregor by the man's height and his long, slightly awkward steps, and from the fact that he was wearing a hat—possibly a baseball cap. McGregor always seemed to be wearing some kind of hat. On Tuesdays there are always several groundskeepers scheduled to work on the grounds, and he would very likely have been scheduled that day.

Select *Assumes to be true* for the statement that describes something the testimony implies the witness assumes to be true, and select *Assumes to be false* for the statement that describes something the testimony implies the witness assumes to be false. Make only two selections, one in each column.

| **Assumes to be true** | **Assumes to be false** | |
|---|---|---|
| | | McGregor was the only groundskeeper scheduled to work at the building on Tuesday, July 6. |
| | | McGregor exited the rear of the building in question alone at 8:00 on the morning of Tuesday, July 6. |
| | | McGregor was scheduled to work at the building on the morning of Tuesday, July 6. |
| | | McGregor's height, way of walking, and tendency to wear hats distinguish him from the other groundskeepers. |
| | | McGregor often wore a baseball cap prior to the morning of Tuesday, July 6. |

•

Ribonucleic acid (RNA) is a molecule built from sequences of smaller molecules called nucleobases. RNA nucleobases are of 4 different types: adenine (A), cytosine (C), guanine (G), and uracil (U). Consider the collection of all possible RNA sequences consisting of 12 nucleobases, 3 of each type. An RNA sequence will be selected at random from this collection, and the first 3 nucleobases of the sequence will be detached from the sequence.

In the table, select the probability that the 3 nucleobases are all of the same type, and select the probability that they are of 3 different types. Make only two selections, one in each column.

| **All the same type** | **3 different types** |
|---|---|
| | $\frac{1}{55}$ |
| | $\frac{9}{220}$ |
| | $\frac{9}{110}$ |
| | $\frac{9}{55}$ |
| | $\frac{27}{55}$ |

•

Let $S$ be the set of all 7-digit numbers $N$ such that the sum of the digits of $N$ is divisible by 7.

$A$ and $B$ are two 7-digit numbers: $A = 4{,}893, \mathrm{P}12 B = 5{,}857{,}0\mathrm{Q}5$

Select a value for $P$ so that $A$ is an element of $S$. Select a value for $Q$ so that $B$ is an element of $S$. Make two selections, one in each column.

| P | Q |
|---|---|

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| ***P*** | ***Q*** | |
|---|---|---|
| | | 4 |
| | | 5 |
| | | 6 |
| | | 7 |
| | | 8 |

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